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Project Title:	Complex Space Radiation-Induced DNA Damage Clusters In Human Cell Transformation: Mechanisms, Relationships And Mitigation		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Comments:	PI Deceased as of October 2009.		
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Predicting and mitigating cancer risk from complex radiation fields in outer space is essential for manned exploration of Mars. This requires understanding the molecular mechanisms by which such radiation inflicts critical damages, responses of cells to radiation, and potentially cancer-inducing changes. Clustered DNA damages are composed of several close-spaced DNA alterations. Specific clusters have different biological impacts, depending on the ability of cells to repair them, and in their cancer-inducing potential. Their levels and types depend on the radiation type, and on factors such as the environment of the DNA. However, repair of clusters in cells exposed to simulated complex radiation fields, the effect of such radiation on induction of potentially oncogenic cellular alterations, and the optimum approaches for mitigating their effects are not known.

The project focuses on four experimental series: 1. Determine if prior exposure of human cells to protons affects repair

Task Description:	of HZE particle-induced double strand breaks, abasic-, oxidized purine- and oxidized pyrimidine clusters; 2. Test whether dual proton-HZE particle exposures affect neoplastic transformation levels of human cells relative to exposure to each radiation species singly; 3. Determine whether levels of slowly-repaired or unrepaired clusters correlate with human cell neoplastic transformation frequencies; and 4. Test whether factors that reduce cluster induction or increase their repair also decrease levels of neoplastic transformation of human cells. Understanding and mitigating space radiation-induced damage to human cells requires understanding the mechanisms of DNA damage induction and cellular repair, and delineating potential relationships between DNA damage clusters and their cancer-incuding potential. These studies should identify approaches for mitigating potentially cancer inducing effects of space radiation exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the role of complex DNA damage clusters in killing and mutating cells can lead to better radiation therapy of cancers. Further, discovering approaches for decreasing DNA damages can minimize the harmful effects of radiation on normal cells.
Task Progress:	This project focused on three critical questions: first, What are the cancer risks in humans from space flight? second, Are biological effects for protons above 10 MeV sufficiently similar to photons that photon data can be used for their consequences? And third, Are there chemopreventive or biological agents that would mitigate acute or late effects? To answer these questions, we used neoplastic transformation of primary human cells, a model for induction of cancer in humans. This system has the advantage of using primary human cells, the same as are found in the human body and are at risk for cancer induction (other models for human cancer induction include the use of either rodents or rodent cells, or human cells that have been treated to make them immortal, unlike those in the human body). These studies were carried out in two locations: irradiations with protons and heavy charged particles are performed at the NASA Space Radiation Laboratory at Brookhaven National Laboratory; X-ray irradiations are carried out in the Radiation Facility at the Medical Department at Brookhaven National Laboratory, and g-ray irradiations are carried out in the Controlled Radiation Environment Facility in the Biology Department at Brookhaven. Human Cell Transformation by Radiation Beams Simulating Deep Space Previous studies of the effects of space radiation have exposed samples to one radiation type, for example only energetic protons or only heavy charged particles. However, although these single ion studies are important for understanding how individual radiation species damage cells, they are not ideal for understanding the actual effects of radiation in deep space, which contains mixed fields of particles, with much higher fluxes of protons than of heavy charged particles (HZE particle). We have first used dual radiation beams to more closely simulate the space radiation environment. Calculations indicate that during a 3-year Mars mission, the nucleus of a cell in an astronaut shielded by 0.4 g/cm2 of Aluminum will experience about 400 proton hits and about 0.4 heavy charged particle hits. Thus most cells will, on average, be hit by a proton before a HZE particle. To simulate cancer risk for human cells exposed to such mixed fields, we therefore tested the effects of dual beam irradiations—both proton exposure before HZE particle exposure and vice versa—as well as the beams individually. During these experiments, we worked closely with the dosimetry physics personnel in the Collider-Accelerator Department (BNL) to develop smooth and rapid methods of switching reliably from one beam to another. The physics staff of the NASA Space Radiation Laboratory also devised a beam imaging charge-coupled device-based camera system, based on the I3 (Integrating Ion Imager), the first charge-coupled device beam imager designed by John Sutherland, BNL, and built by John Trunk of Dr. Sutherland's group. These cameras allow rapid checking of the beam uniformity to assure even and uniform irradiation of all cells in a sample. Using this system, we developed methods for switching between proton and HZE beams within about 2 minutes, and for simultaneous checking of the uniformity of the beam. The systems allow real-time imaging of the cell samples in the beam, thus insuring that the irradiation of the samples is correct. This is essential since after irradiation, at least 4 week's work is required to obtain the survival and transformation results. Thus in addition to saving beam time (and thereby maximizing scientific output per cost to NASA), the beam imaging systems provide validation of the irradiations, which is essential for interpretation of the biological results. Using the beam imaging system and the protons and heavy ion beams at the NASA Space Radiation Laboratory, we found that the dual irradiations (protons plus heavy charged particle) can produce about three times as much transformation than the sum of the single irradiations (Zhou et al, Radiation Research, 2006). At longer spacings, the transformation level is just that expected from the sum of the two irradiations (Sutherland et al., Radiation Research, 2005). The timing between the irradiations is critical, with a window of about 3 hr giving the three-fold increase in transformation. The increase in transformation is not limited to one type of HZE particle, with both the proton-iron ion and the proton-titanium ion dual beam irradiations giving a similar 3-fold increase in transformation. Further, for both proton-HZE dual irradiations, the order of the beams is critical, with protons following HZE particles giving about the same transformation level as the sum of the two beams given separately. Given some basic assumptions and conditions (first, the expected radiation exposures during space travel; second, the 10(14) cells in a human body; third, 3-hr critical window for susceptibility to increased transformation; and fourth, the shielding of 0.4 g/cm2 of aluminum) we can calculate that ~ 2 x 10(12) cells (~ 2% of the total) in a space traveler would be at risk for increased probability of transformation. It thus becomes essential to develop effective countermeasures against transformation, rather than having a goal of protecting only against radiation-induced cell killing. We also tested the effectiveness of protons and HZE particles in inducing cell transformation. We irradiated cells with 290 MeV/n C, 600 MeV/n Si, 1 GeV/n Ti, or 1 GeV/n Fe ions; the efficiency of transformation is a linear function of the ion's linear energy transfer value. The transformation efficiency also increased with increasing atomic number of the ion (Z). Protons Produce Different DNA Damages than X and gamma-rays An important question in evaluating space radiation risk is whether the biological effects of high energy protons are similar to those of X and gamma-rays so that the extensive literature on X- and gamma-rays can be used to evaluate human risk. It had been assumed that protons, X- and gamma-rays would have similar effects, because they all deposit

	similar amounts of energy as they traverse matter. However, protons have different patterns of energy deposition due to their track structure. Measurements in a simple DNA model system indicates that protons damage DNA in a similar manner to other charged particles (like iron ions and titanium ions), and quite different from X- and gamma-rays (Hada and Sutherland, Radiation Research, 2006). Testing of human cells for DNA damage induction and transformation by protons and photons is under way. Countermeasures Against Radiation-induced Cell Transformation To identify effective countermeasures that could be rapidly to protect astronaut crews against space radiation damage, we focused on substances naturally found in the human body and/or currently available for human consumption. Melatonin—a human hormone-- meets these criteria, and is a good candidate because it is an excellent scavenger of hydroxyl radicals, which are important in producing DNA damage. Our data show that melatonin provides excellent protection against cell transformation produced by a hit of a HZE particle, with little effect on cell survival.
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